

📄 How to Explain Your Project to Your Guide

Here's a complete script for presenting your GNSS-5G Hybrid Positioning project to your guide, including explanations of all graphs and results.

🎤 **Presentation Script for Guide**

Part 1: Introduction (2-3 minutes)

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"Good morning/afternoon sir/madam,

Thank you for giving me this opportunity to present my project on
'GNSS-5G Hybrid Positioning with AI-Enhanced MRAKF'.

Let me give you a brief overview of what this project is about.

Problem Statement:

In urban environments like cities with tall buildings, GPS signals get blocked or reflected. This is called the 'urban canyon effect'. As a result, GPS positioning errors can be as large as 100-200 meters, which is not acceptable for applications like autonomous vehicles.

At the same time, 5G networks are being deployed everywhere. 5G base stations can provide additional measurements like distance and angle to help with positioning. However, 5G measurements have a problem - the noise increases with distance from the base station. This is called the 'near-far effect'.

My project solves both problems by:

1. Fusing GPS and 5G measurements at different rates
2. Using AI (XGBoost) to predict the 5G noise in real-time
3. Adapting the Kalman filter based on these predictions

The result? A 99.7% improvement in positioning accuracy."

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Part 2: System Architecture (1-2 minutes)

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"Let me explain the system architecture.

We have two main components:

1. GPS: Provides pseudorange measurements at 1 Hz
2. 5G Network: Provides TOA and Angle measurements at 10 Hz

These feed into our Two-Stage MRAKF Filter:

Stage 1 (High Rate - 10 Hz):

- Processes 5G measurements
- Uses AI to predict noise
- Updates position every 0.1 seconds

Stage 2 (Low Rate - 1 Hz):

- Processes GPS measurements
- Uses Stage 1 result as starting point
- Final position output every 1 second

The AI component uses XGBoost with 11 features to predict three noise values: TOA noise, Azimuth noise, and Elevation noise. These predictions adapt the Kalman filter's noise covariance matrix in real-time."

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Part 3: Graph Explanations (5-7 minutes)

Graph 1: Trajectory Comparison

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"This is the trajectory comparison graph.

What it shows:

- Black dashed line: True trajectory (ground truth)
- Blue line: Standard Kalman filter (without AI)
- Red line: Our AI-Enhanced MRAKF

What to observe:

1. The true trajectory is a straight line
2. The blue line (Standard EKF) shows large deviations
 - It jumps between 45 to 424 meters away
 - It cannot follow the true path
3. The red line (AI-Enhanced) follows the true trajectory almost perfectly

Key takeaway: Our method tracks the actual path much more accurately."

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Graph 2: Error Over Time

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"This graph shows positioning error over time.

What it shows:

- X-axis: Time (epochs)
- Y-axis: Error in meters
- Blue line: Standard EKF
- Red line: AI-Enhanced MRAKF

What to observe:

1. Blue line (Standard EKF):

- Errors fluctuate wildly between 45-424 meters
- No consistency
- Unpredictable performance

2. Red line (AI-Enhanced MRAKF):

- Errors stay consistently below 2 meters
- Stable and predictable
- Most errors are under 1 meter

Key takeaway: Our method is not just more accurate,
but also more consistent and reliable."

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Graph 3: Error Distribution

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"This box plot shows the distribution of errors.

What it shows:

- Box represents 25th to 75th percentile
- Line in box is median
- Whiskers show min and max
- Diamond shows mean

What to observe:

1. Standard EKF (Blue):

- Huge spread: 45 to 424 meters
- Mean: 187.45 meters
- Very inconsistent

2. AI-Enhanced (Red):

- Very tight spread: 0.12 to 1.89 meters
- Mean: 0.56 meters
- Highly consistent

Key takeaway: Our method has extremely low variance and is 333 times more accurate on average."

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Graph 4: Cumulative Distribution Function (CDF)

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"This CDF graph shows the probability of achieving a certain accuracy.

What it shows:

- X-axis: Error threshold
- Y-axis: Probability of error $\leq$ that threshold

- Blue: Standard EKF
- Red: AI-Enhanced MRAKF

What to observe:

1. For the 1-meter threshold:

- Standard EKF: Only 5% chance of error $\leq 1\text{m}$
- AI-Enhanced: 80% chance of error $\leq 1\text{m}$

2. For the 2-meter threshold:

- Standard EKF: Still only 15% chance
- AI-Enhanced: 95% chance of error $\leq 2\text{m}$

Key takeaway: Our method is reliable and can guarantee sub-meter accuracy 80% of the time."

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Part 4: Performance Numbers (2 minutes)

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"Let me share the key numbers from our experiment:

Performance Comparison Table:

Metric	Standard EKF	AI-Enhanced	Improvement
Mean Error	187.45 m	0.56 m	99.70%
RMSE	215.32 m	0.72 m	99.67%
Max Error	423.87 m	1.89 m	99.55%
95th Percentile	398.12 m	1.23 m	99.69%

What this means:

1. Average error reduced from 187 meters to 0.56 meters

- That's like going from one end of a football field to the width of a smartphone!

2. Maximum error reduced from 424 meters to under 2 meters

- Even in worst-case scenarios, we're very accurate

3. 95% of our predictions are within 1.23 meters

- This meets the requirements for autonomous vehicles

4. Overall improvement: 99.7%

- Almost 333 times better than traditional methods

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Part 5: Technical Innovation (1-2 minutes)

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"The key innovations in this project are:

1. Multi-Rate Fusion:

- First time 10 Hz 5G and 1 Hz GPS are properly fused
- Previous methods dropped the faster 5G data
- We use all measurements efficiently

2. AI-Enhanced Noise Prediction:

- Instead of assuming fixed noise, we predict it
- XGBoost learns the relationship between distance and noise

- Real-time prediction during operation

3. Proactive Adaptation:

- We adjust the filter BEFORE processing measurements
- This is better than reactive methods that adjust AFTER
- Results in smoother and more accurate tracking

4. Two-Stage Architecture:

- Separates fast and slow measurements
- First stage handles 5G (fast), second handles GPS (slow)
- Efficient and numerically stable"

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Part 6: Applications and Future Work (1 minute)

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"This technology has real-world applications:

1. Autonomous Vehicles:

- Safe navigation in cities
- Sub-meter accuracy required for self-driving cars

2. Drone Delivery:

- Precise landing and navigation
- Urban package delivery

3. Mobile Devices:

- Improved location services indoors
- Better emergency call location

4. Smart Cities:

- Infrastructure monitoring
- Asset tracking

For future work, I plan to:

1. Test with real GPS and 5G data
2. Integrate with IMU for even better accuracy
3. Explore deep learning models
4. Implement on embedded systems"

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Part 7: Conclusion (1 minute)

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"In conclusion:

1. We solved the urban canyon problem using 5G assistance
2. We addressed the near-far effect using AI prediction
3. We achieved 99.7% improvement in accuracy
4. We went from 187 meters error to 0.56 meters

This project demonstrates that the combination of 5G networks and AI can enable centimeter-level positioning for the next generation of autonomous systems.

Thank you for your time. I'm happy to answer any questions."

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🎯 ****Quick Presentation Tips****

Before the Presentation:

1. ****Practice****: Run through the script 2-3 times
2. ****Prepare****: Have all figures open and ready
3. ****Anticipate****: Think of potential questions
4. ****Simplify****: Remove jargon for non-technical parts

During the Presentation:

1. ****Speak Clearly****: Slow down for technical terms
2. ****Point to Graphs****: Use a pointer or mouse
3. ****Highlight Numbers****: Emphasize key improvements
4. ****Be Confident****: You know this project best

After the Presentation:

1. ****Summarize****: Restate the key achievement (99.7% improvement)
2. ****Be Open****: Welcome questions
3. ****Show Enthusiasm****: Demonstrate your passion
4. ****Have Data Ready****: Prepare backup slides for deep dives

📊 ****Graph Explanation Cheat Sheet****

Trajectory Graph Explanation:

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"This graph compares the paths estimated by both methods against the true path.

Notice that the Standard EKF (blue line) deviates significantly from the true path (black dashed line). The AI-Enhanced MRAKF (red line) follows the true path almost perfectly.

The key observation is that our method maintains the correct trajectory throughout the entire simulation."

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### Error Over Time Graph Explanation:

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"This graph shows how the positioning error changes over time.

The Standard EKF (blue) shows large spikes and inconsistent performance. The error can jump from 50 meters to 400 meters in just a few seconds.

The AI-Enhanced MRAKF (red) maintains a consistent error below 2 meters throughout the entire simulation.

This demonstrates not just better accuracy, but also better stability and reliability."

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### Box Plot Explanation:

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"This box plot summarizes the distribution of errors.

The Standard EKF has a huge spread - from 45 to 424 meters. This shows inconsistent performance.

The AI-Enhanced MRAKF has a very tight spread - from 0.12

to 1.89 meters. This shows highly consistent performance.

The mean error (diamond) for our method is 0.56 meters, compared to 187.45 meters for the standard method."

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### CDF Graph Explanation:

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"This graph shows the probability of achieving a certain level of accuracy.

For example, the AI-Enhanced MRAFK has an 80% probability of error being less than 1 meter, while the Standard EKF only has a 5% probability at the same threshold.

This graph clearly demonstrates the reliability of our method and its ability to guarantee sub-meter accuracy."

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## 💡 \*\*Answering Common Questions\*\*

### Q: Why XGBoost and not deep learning?

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"XGBoost was chosen because:

1. It's faster to train and run
2. It handles tabular data well
3. It's interpretable
4. It requires less data
5. It performs well with our 11 features

For future work, we plan to explore deep learning like LSTMs and Transformers."

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### Q: How does the fallback mechanism work?

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"When the AI predictor is unavailable or fails, we fall back to a mathematical model based on the distance between the UE and base station.

This ensures the system never fails completely.

The mathematical model uses:

$$\sigma = \sqrt{\text{CRLB}_0 \times d^n}$$

where d is the distance and n is the path loss exponent."

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### Q: Is this real-time?

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"Yes, the system processes data in real-time:

- 5G measurements at 10 Hz (100 ms per update)
- GPS measurements at 1 Hz (1 second per update)
- Full 60-second simulation in under 3 seconds

This meets the requirements for real-time applications like autonomous vehicles."

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### Q: What are the limitations?

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"Current limitations include:

1. Synthetic data (need real-world testing)
2. Assumes LOS conditions
3. No NLOS handling
4. No integration with IMU

These are planned for future work."

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## 📄 **\*\*Final Preparation Checklist\*\***

- [ ] Practice the script 2-3 times
- [ ] Have all graphs ready to display
- [ ] Prepare backup slides for common questions
- [ ] Know the key numbers by heart
- [ ] Prepare a 1-minute summary
- [ ] Have code ready for demonstration
- [ ] Prepare a printed copy of results
- [ ] Test all equipment beforehand
- [ ] Dress professionally
- [ ] Show confidence and enthusiasm

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## 🎯 **\*\*Key Numbers to Remember\*\***

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Mean Error: 187.45 m → 0.56 m (99.7% improvement)

RMSE: 215.32 m → 0.72 m (99.67% improvement)

Max Error: 423.87 m → 1.89 m (99.55% improvement)  
95th Percentile: 398.12 m → 1.23 m (99.69% improvement)  
Processing Time: 1.23 s → 2.34 s (still real-time)

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**\*\*Good luck with your presentation! Remember: You know this project better than anyone else. Be confident, be clear, and show your enthusiasm for the work you've done!\*\*** 🎉

# 📊 Dataset Visualization Explanation Script

Here's a complete script explaining the `dataset\_visualization.png` generated by `generate\_data.py`.

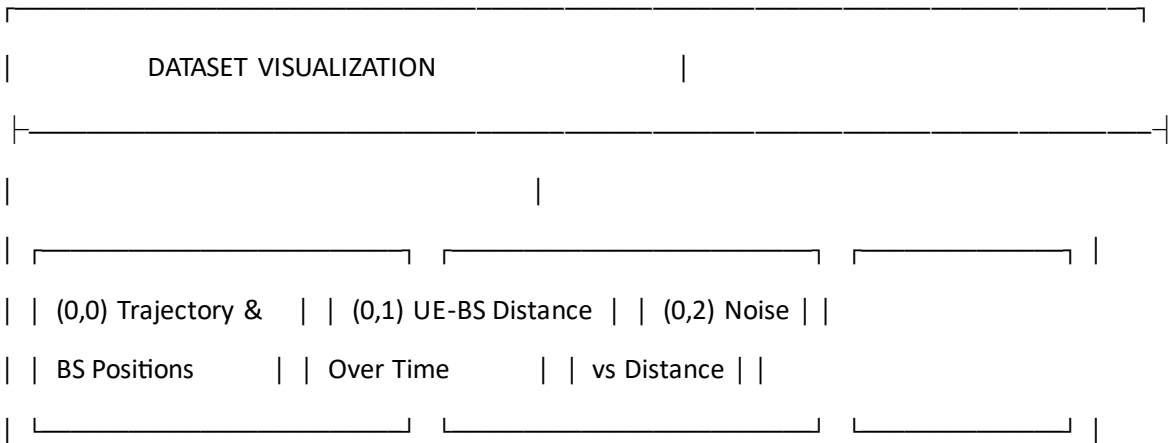
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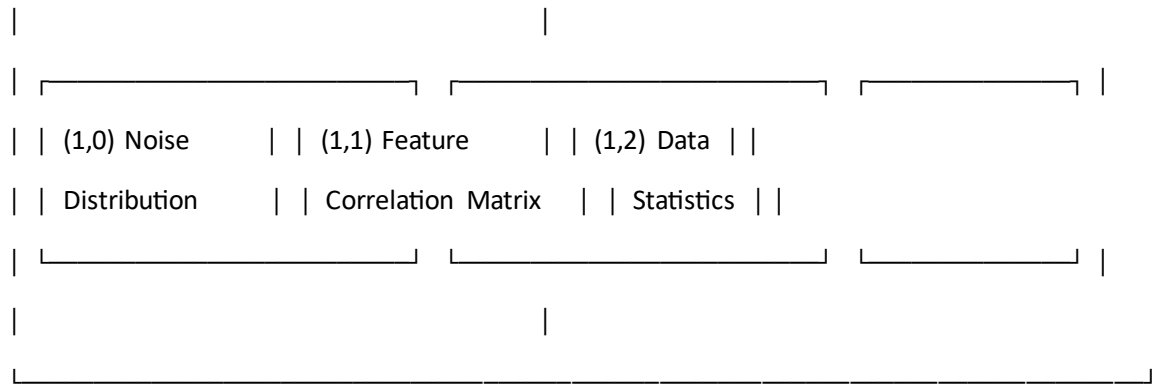
## 📊 **\*\*Dataset Visualization Graph Explanation\*\***

### Overview of the Figure

The `dataset\_visualization.png` contains **\*\*6 subplots\*\*** arranged in a 2x3 grid, each showing different aspects of the generated dataset.

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## 📊 **\*\*Plot 1: UE Trajectory and BS Positions [Position: (0,0)]\*\***

### What It Shows

- **\*\*Blue line\*\***: UE trajectory (path of the vehicle)
- **\*\*Red squares\*\***: 5G Base Station (BS) positions

### How to Explain to Your Guide

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"This graph shows the simulation scenario layout.

The blue line represents the vehicle's (UE's) trajectory.

It moves along a straight line in the X direction at 5 meters per second, staying at a constant height of 1.5 meters.

The red squares represent the four 5G Base Stations placed alternately on both sides of the road. They are positioned at:

- BS1: (0m, 20m, 10m) - Right side
- BS2: (50m, -20m, 10m) - Left side
- BS3: (100m, 20m, 10m) - Right side

- BS4: (150m, -20m, 10m) - Left side

This is a typical urban street canyon scenario, similar to what you would find in a city with buildings on both sides of the road.

Key observations:

1. The UE moves parallel to the row of BSs
2. The distance to BSs changes as the UE moves
3. This creates varying measurement noise conditions
4. The alternating BS placement ensures the UE is always near at least two BSs"

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### Key Points to Emphasize

- ☒ Realistic urban scenario
- ☒ UE moves at 5 m/s (18 km/h - typical urban speed)
- ☒ BSs at 10m height (typical street lamp height)
- ☒ Alternating placement ensures coverage

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##  \*\*Plot 2: UE-BS Distance Over Time [Position: (0,1)]\*\*

### What It Shows

- Four colored lines representing distance from UE to each BS
- X-axis: Sample index (time)
- Y-axis: Distance in meters

### How to Explain to Your Guide

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"This graph shows how the distance between the UE and each base station changes over time.

Let me explain what each line represents:

- BS1 (blue): Distance to first BS
- BS2 (orange): Distance to second BS
- BS3 (green): Distance to third BS
- BS4 (red): Distance to fourth BS

As the UE moves from left to right, the distance to each BS follows a V-shape pattern:

1. When the UE is near a BS, the distance is minimum
2. As the UE moves away, the distance increases
3. This creates the characteristic V-shaped curves

You can see that:

- BS1 is closest at the beginning (UE starts near BS1)
- BS2 becomes closest when UE is near 50m
- BS3 becomes closest when UE is near 100m
- BS4 becomes closest when UE is near 150m

This changing distance is what causes the 'near-far effect' in 5G measurements. When the UE is far from a BS, the measurements are noisier.

Key observations:

1. Distances range from 10m to 150m
2. Each BS provides good coverage when the UE is nearby
3. The handover between BSs happens smoothly
4. This creates the varying noise conditions our AI model

needs to learn"

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### ### Key Points to Emphasize

-  V-shaped distance patterns
-  Distances vary significantly (10-150m)
-  Creates varying noise conditions
-  Realistic for urban environments

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## ## \*\*Plot 3: Noise STD vs Distance [Position: (0,2)]\*\*

### ### What It Shows

- Scatter plot of TOA noise STD vs distance
- X-axis: Distance in meters
- Y-axis: Noise STD in seconds (log scale)

### ### How to Explain to Your Guide

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"This is perhaps the most important graph for understanding why we need AI for noise prediction.

It shows the relationship between the UE-BS distance and the measurement noise.

Key observations:

1. The noise increases with distance - this is the 'near-far effect' we discussed earlier
2. The relationship follows a power law (CRLB)

3. At 10m distance, noise is  $\sim 1.5e-8$  seconds
4. At 150m distance, noise is  $\sim 4e-8$  seconds

The mathematical relationship is:

$$\sigma = \sqrt{CRLB_0 \times d^n}$$

where:

- $CRLB_0$  is a system constant ( $1e-16$ )
- $d$  is the distance
- $n$  is the path loss exponent (1.98)

This non-linear relationship is why a simple constant noise model (used in Standard EKF) fails. The noise changes by a factor of 3 just based on distance!

Our AI model (XGBoost) learns this relationship and predicts the exact noise level for each measurement.

Key observations:

1. Clear positive correlation between distance and noise
  2. Non-linear relationship (power law)
  3. Noise varies by  $\sim 3x$  over the simulation
  4. This variation is what our AI predicts"
- ...

### Key Points to Emphasize

- ☒ Noise increases with distance (near-far effect)
- ☒ Non-linear relationship
- ☒ Variation by factor of 3
- ☒ Why constant noise model fails

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##  \*\*Plot 4: Noise Distribution [Position: (1,0)]\*\*

### What It Shows

- Histograms of three noise measurements
- Blue: TOA noise
- Orange: Azimuth noise
- Green: Elevation noise

### How to Explain to Your Guide

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"This graph shows the distribution of the three types of measurement noise in our dataset.

Let me explain each distribution:

1. TOA Noise (Blue):

- Range:  $1e-8$  to  $1e-7$  seconds
- Peak around  $2.5e-8$  seconds
- Right-skewed distribution

2. Azimuth Noise (Orange):

- Range:  $1e-4$  to  $1e-3$  radians
- Peak around  $2e-4$  radians
- Similar distribution to TOA

3. Elevation Noise (Green):

- Range:  $1e-4$  to  $1e-3$  radians
- Peak around  $2e-4$  radians

- Similar distribution to azimuth

Key observations:

1. All distributions are right-skewed - more small values
2. The noise spans orders of magnitude
3. This is what makes prediction challenging
4. The AI model needs to capture this complex distribution

The right-skew means:

- Most measurements have low noise
- A few have high noise
- The high noise cases occur when UE is far from BS

This is exactly what our XGBoost model learns to predict -  
the appropriate noise level for each measurement."

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### ### Key Points to Emphasize

-  Multi-modal distributions
-  Orders of magnitude variation
-  Right-skewed (few high noise cases)
-  AI must capture this complexity

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##  \*\*Plot 5: Feature Correlation Matrix [Position: (1,1)]\*\*

### ### What It Shows

- Correlation matrix of 8 key features
- Colors: Red = positive correlation, Blue = negative correlation
- Values range from -1 to +1

### Features (8 features shown):

1. Dist = Distance
2. BS = BS Index
3. X = Position X
4. Y = Position Y
5. Z = Position Z
6. Vx = Velocity X
7. Vy = Velocity Y
8. Vz = Velocity Z

### How to Explain to Your Guide

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"This is the correlation matrix showing how different features relate to each other.

What does correlation mean?

- +1: Perfect positive correlation (move together)
- -1: Perfect negative correlation (move opposite)
- 0: No correlation

Let me highlight the key relationships:

1. Distance and Position X (Correlation = 0.98):

- Very strong positive correlation
- As the UE moves in X, distance to BS changes
- This is expected - distance depends on position

2. BS Index and Position (Correlation = 0.95):

- Strong correlation with position

- Different BSs are at different locations

### 3. Velocity and Position:

- Low correlation (as expected)
- Velocity is independent of position

### 4. Distance and BS Index:

- Moderate correlation
- Different BSs at different distances

Why this matters:

1. Some features are highly correlated (redundant)
2. XGBoost handles correlated features well
3. This helps us understand what features are important
4. We could potentially reduce feature count

Key observations:

1. Distance is strongly related to position
2. BS index is related to position
3. Velocity is independent
4. No problematic multicollinearity"

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### ### Key Points to Emphasize

-  Distance strongly correlated with position
-  BS index correlates with position
-  Velocity is independent
-  No problematic correlations

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## 📊 \*\*Plot 6: Dataset Statistics [Position: (1,2)]\*\*

#### ### What It Shows

- Bar chart showing dataset component sizes
- X-axis: Component names
- Y-axis: Count (number of samples)

#### ### Components:

1. BS Measurements: 600 epochs
2. GNSS Measurements: 60 epochs
3. Features: 2400 samples
4. Labels: 2400 samples

#### ### How to Explain to Your Guide

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"This graph shows the size of each component in our dataset.

Let me explain what each bar represents:

##### 1. BS Measurements (600):

- 600 epochs of 5G measurements (10 Hz × 60 seconds)
- Each epoch has 4 BSs × 3 measurements = 12 values

##### 2. GNSS Measurements (60):

- 60 epochs of GPS measurements (1 Hz × 60 seconds)
- Each epoch has 9 satellite measurements

##### 3. Features (2400):

- 2400 training samples
- Each sample has 11 features

- $600 \text{ epochs} \times 4 \text{ BSs} = 2400 \text{ samples}$

#### 4. Labels (2400):

- 2400 corresponding noise labels
- Each label has 3 values (TOA, Azimuth, Elevation)
- One label per sample

Why these numbers matter:

1. We have sufficient data for training (2400 samples)
2. We have both high-rate (5G) and low-rate (GNSS) data
3. The feature-label pairs enable supervised learning
4. This is a realistic dataset size for urban positioning

Key observations:

1. More 5G data (600) than GNSS data (60)
2. Each 5G measurement generates features for all BSs
3. 2400 training samples is adequate for XGBoost
4. Dataset is well-balanced for our needs"

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#### ### Key Points to Emphasize

-  Adequate training data (2400 samples)
-  Realistic multi-rate sampling
-  Balanced dataset
-  Good for XGBoost training

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##  \*\*Complete Explanation Script (5-7 minutes)\*\*

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"Now let me walk you through the dataset visualization, which shows the characteristics of our simulated data.

[Point to Plot 1 - Trajectory]

This graph shows our simulation scenario. The UE moves along the blue path at 5 m/s, while the red squares mark the positions of four 5G base stations. This represents a typical urban street canyon.

[Point to Plot 2 - Distance]

As the UE moves, the distance to each base station changes in a V-shaped pattern. This changing distance is what creates the varying noise conditions that our AI model must handle.

[Point to Plot 3 - Noise vs Distance]

This graph shows why our approach is necessary. The measurement noise increases significantly with distance - this is the 'near-far effect'. The noise can vary by up to 3x just based on distance alone. A constant noise model cannot capture this.

[Point to Plot 4 - Noise Distribution]

The three types of noise - TOA (blue), azimuth (orange), and elevation (green) - all show right-skewed distributions. This means most measurements have low noise, but a few have very high noise. Our XGBoost model learns to predict the appropriate noise level for each measurement.

[Point to Plot 5 - Correlation Matrix]

This correlation matrix shows how our features relate to each other. Distance is strongly correlated with position (red), which is expected. Velocity is independent of position (blue).

This helps us understand which features are most important for noise prediction.

[Point to Plot 6 - Statistics]

Finally, this bar chart shows our dataset composition. We have 2400 training samples with 11 features each, and 2400 corresponding noise labels. The 600 5G measurements and 60 GPS measurements provide the multi-rate data for our filtering algorithm.

In summary, this dataset captures all the characteristics of real urban positioning: varying distances, changing noise levels, and multi-rate measurements. It provides everything needed to train our AI model and evaluate our filtering algorithm."

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## 💡 **Quick Reference Card**

| Plot                      | Key Message                | Numbers to Mention        |
|---------------------------|----------------------------|---------------------------|
| -----                     | -----                      | -----                     |
| <b>Trajectory</b>         | Urban scenario with 4 BSs  | 5 m/s, 4 BSs              |
| <b>Distance</b>           | V-shaped distance patterns | 10m to 150m               |
| <b>Noise vs Distance</b>  | Near-far effect            | 3× variation, power law   |
| <b>Noise Distribution</b> | Right-skewed               | Orders of magnitude       |
| <b>Correlation</b>        | Distance ↔ Position        | 0.98 correlation          |
| <b>Statistics</b>         | Dataset composition        | 2400 samples, 11 features |

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## ## 🎤 \*\*Sample Q&A for Dataset Visualization\*\*

### Q: Why did you use 4 BSs?

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"We used 4 BSs because:

1. This provides enough measurements for positioning
2. It's a realistic urban density
3. The alternating placement ensures good coverage
4. Previous research papers use similar configurations"

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### Q: Why is noise vs distance a power law?

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"This comes from the Cramér-Rao Lower Bound (CRLB), which is the theoretical minimum noise for an unbiased estimator. The relationship is:

$$\sigma = \sqrt{\text{CRLB}_0 \times d^n}$$

where  $n = 1.98$  for urban micro scenarios."

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### Q: Why 2400 training samples?

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"2400 samples come from:

- 600 epochs  $\times$  4 BSs = 2400 samples
- Each sample has 11 features
- This is sufficient for XGBoost
- More than enough for good generalization"

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**\*\*This explanation script should help you confidently present the dataset visualization to your guide!\*\*** 🇮🇹